SimpleScore User Manual

1. Installation

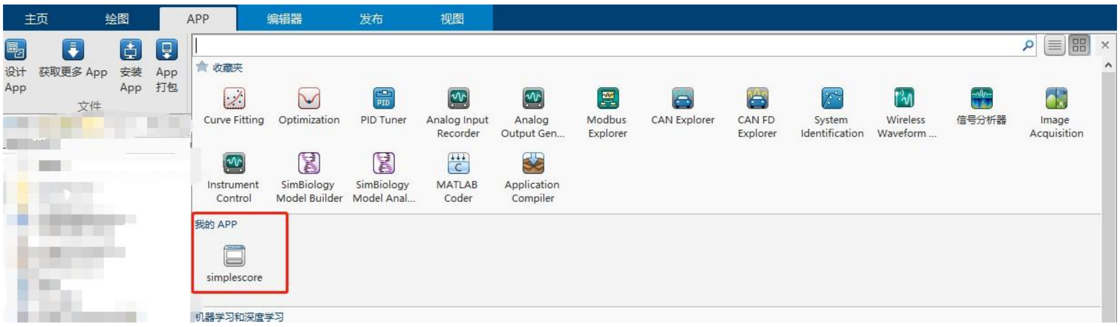
Open **MATLAB** (recommended version **R2021a or later**).

Open the **APP** tab in the MATLAB menu bar and select **Install App**.

Choose the provided file **simplescore.mlappinstall** to install the application.

After installation is complete, the app

will appear under **My Apps** in the APP menu.

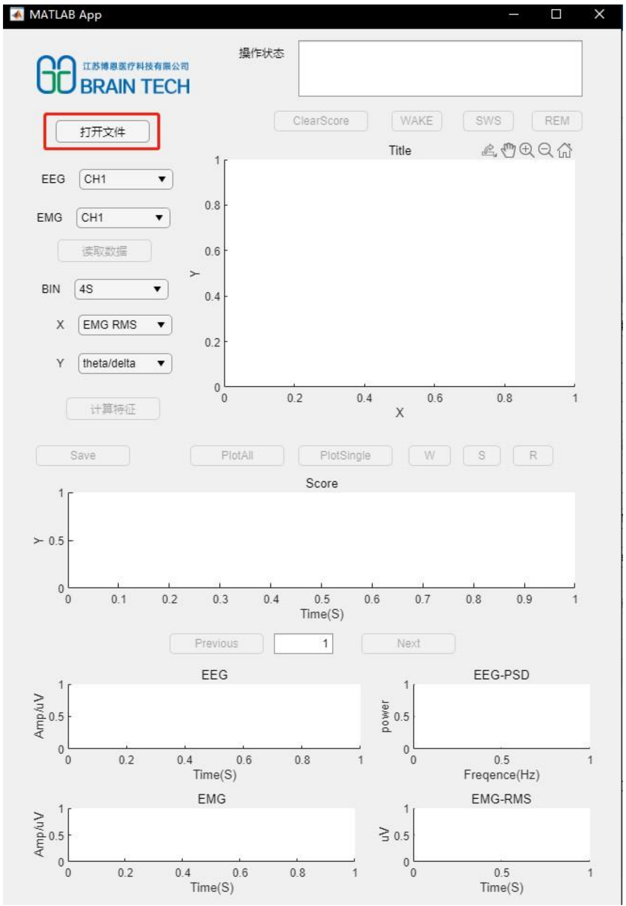


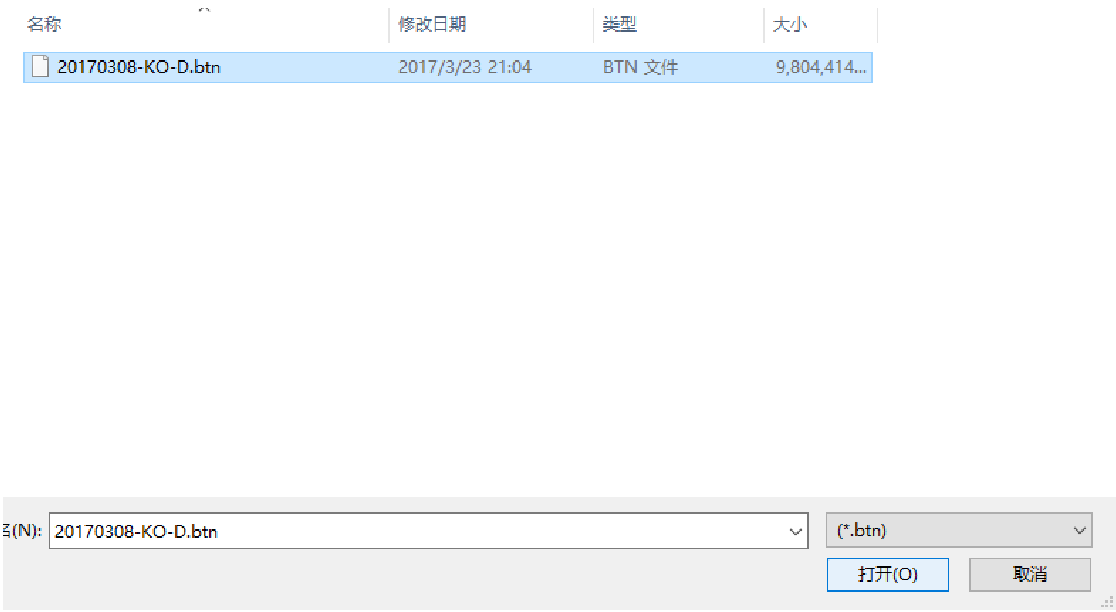
2. Usage

2.1 Load Data

Open the software and click **[Open File]** to select the data file to be analyzed.

During file loading, the button will appear **dark gray**, indicating that the operation is in progress.





If no file is selected or if the file format is incorrect, the message **"No File"** will be displayed

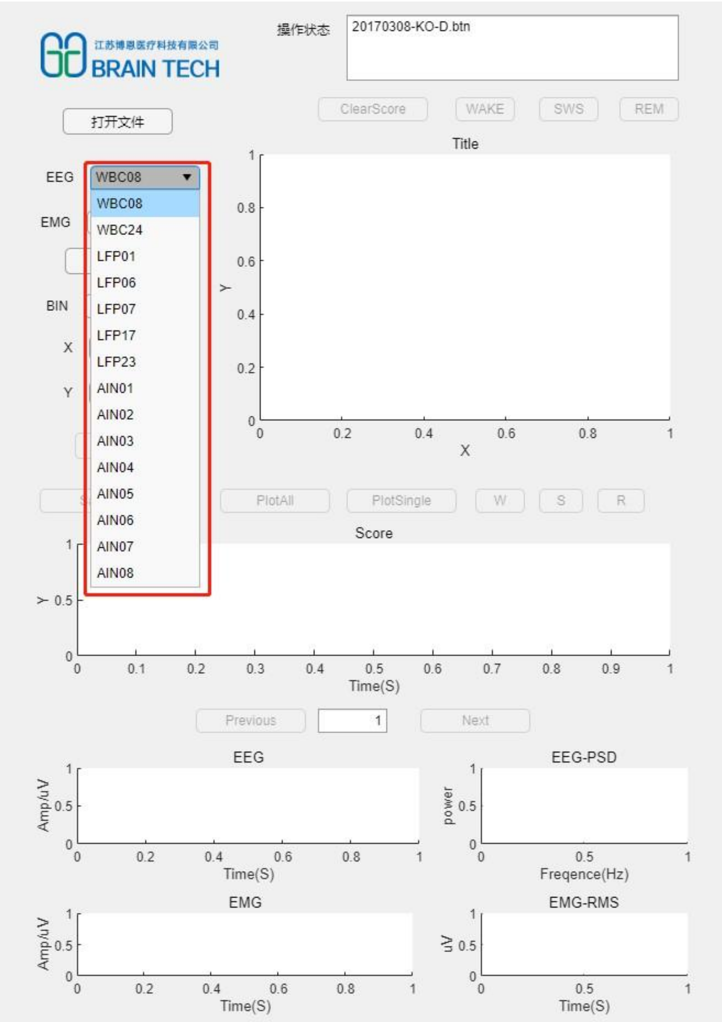
2.2 Channel Selection

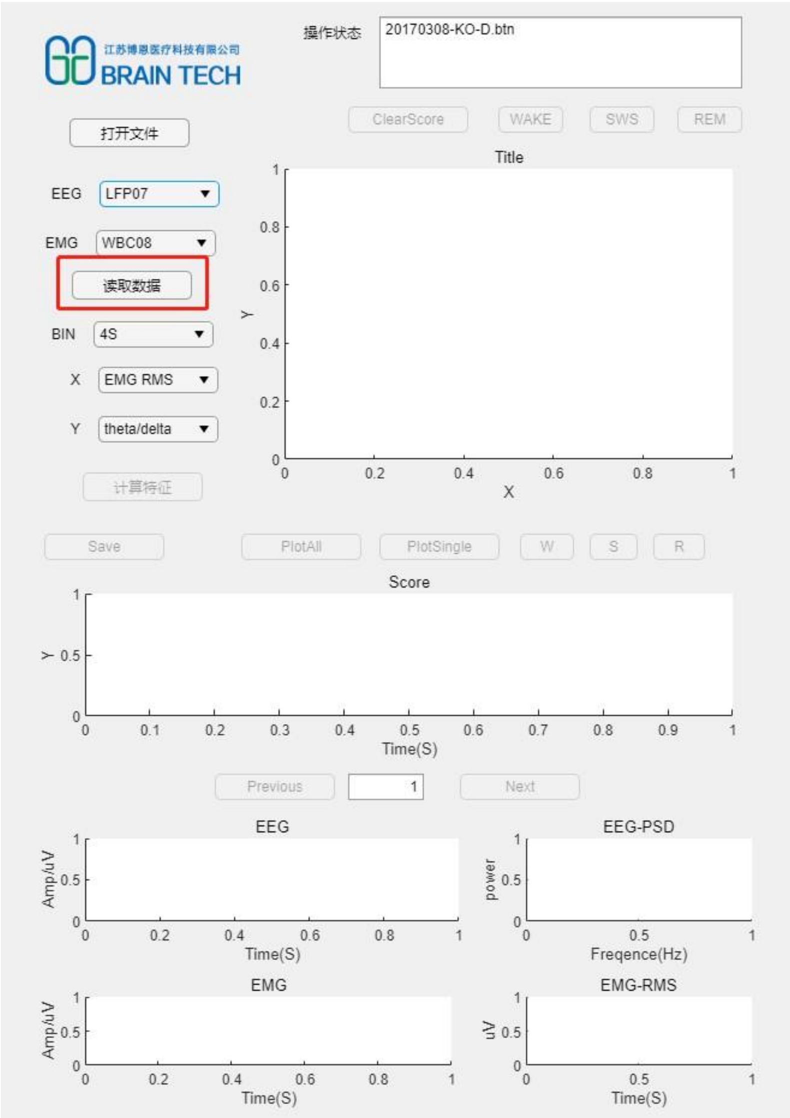
Select the **EEG** and **EMG** channels from the drop-down menus.

Click **[Read Data]** to load the selected channels.

During data reading, the button will appear **dark gray**.

Once reading is complete, the message **"read over"** will be displayed.



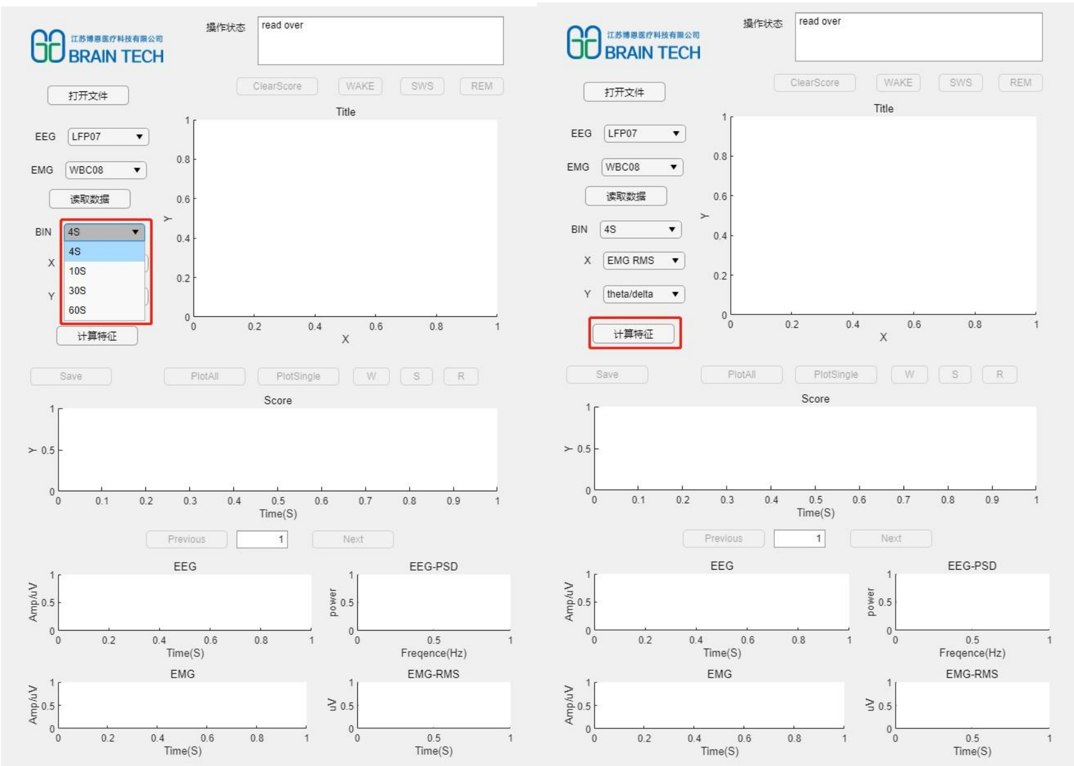


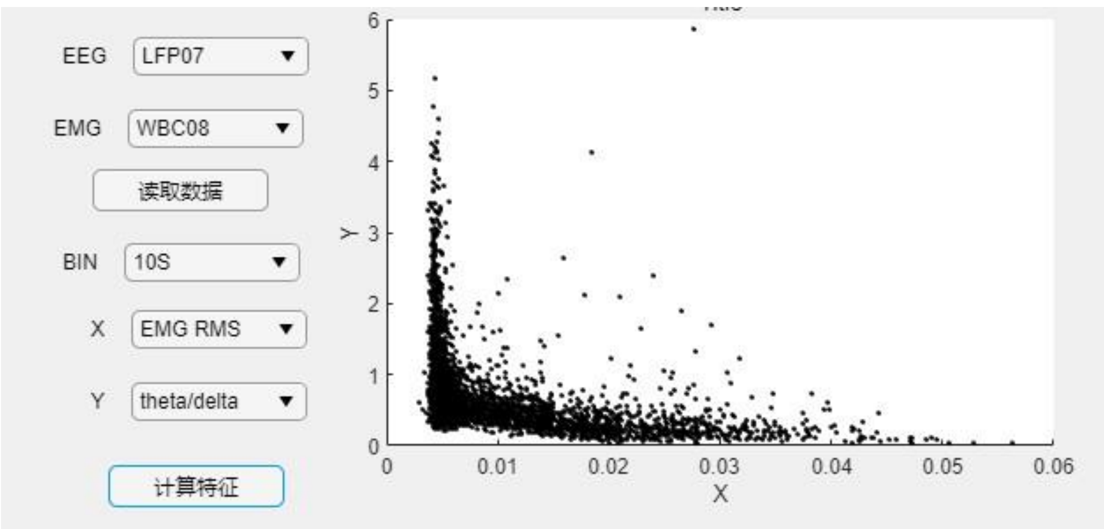
2.3 Feature Calculation

Select the **time window size**, then click **[Calculate Features]** to segment the data and compute feature values.

During computation, the button will appear **dark gray**.

After completion, the calculated features will be displayed and plotted automatically.



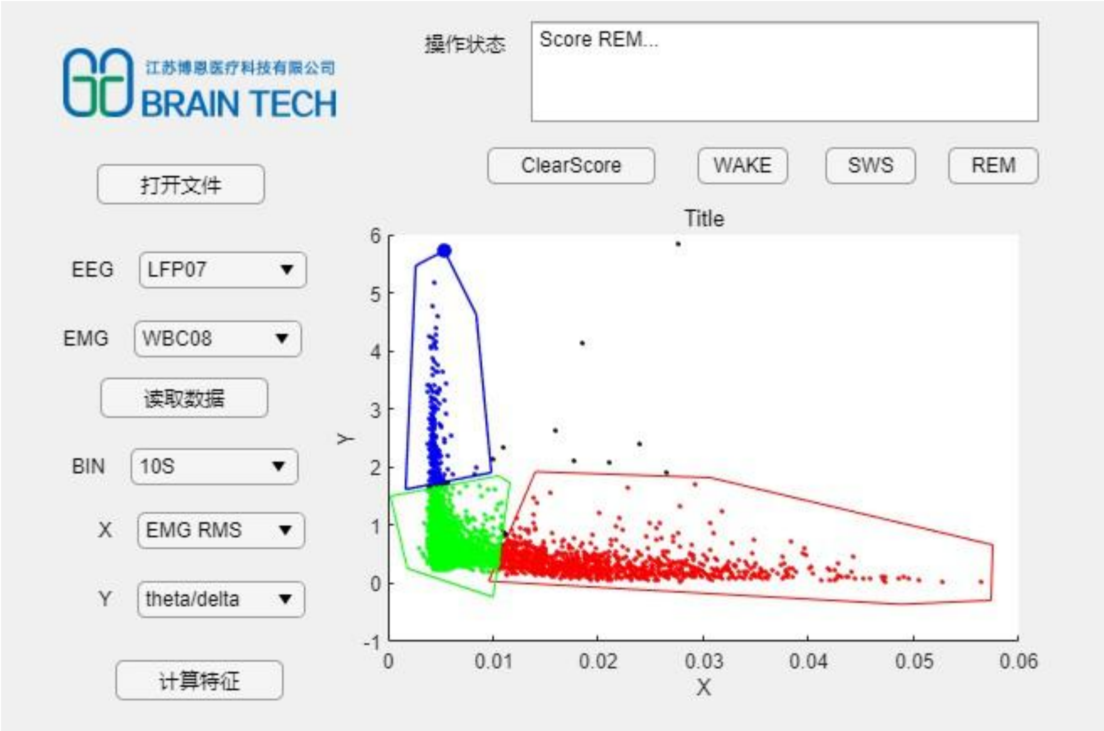


2.4 Sleep Stage Scoring

Select **[WAKE] / [SWS] / [REM]** for manual sleep stage scoring.

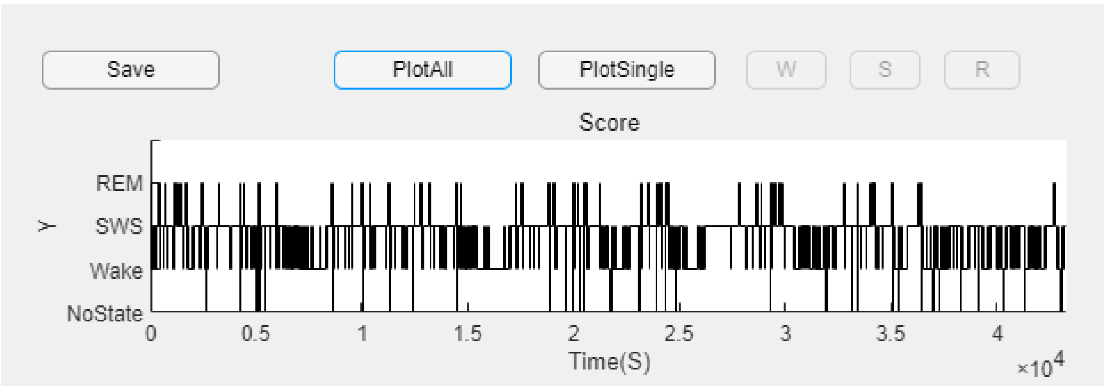
- **Left mouse click**: select the boundary of a stage
- **Right mouse click**: assign the sleep stage label

If the manual scoring result is unsatisfactory, click **[ClearScore]** to clear the current classification and re-score.



2.5 Sleep Stage Visualization

Click **[PlotAll]** to plot sleep staging results for the **entire recording duration**.



2.6 Sleep Stage Fine Adjustment

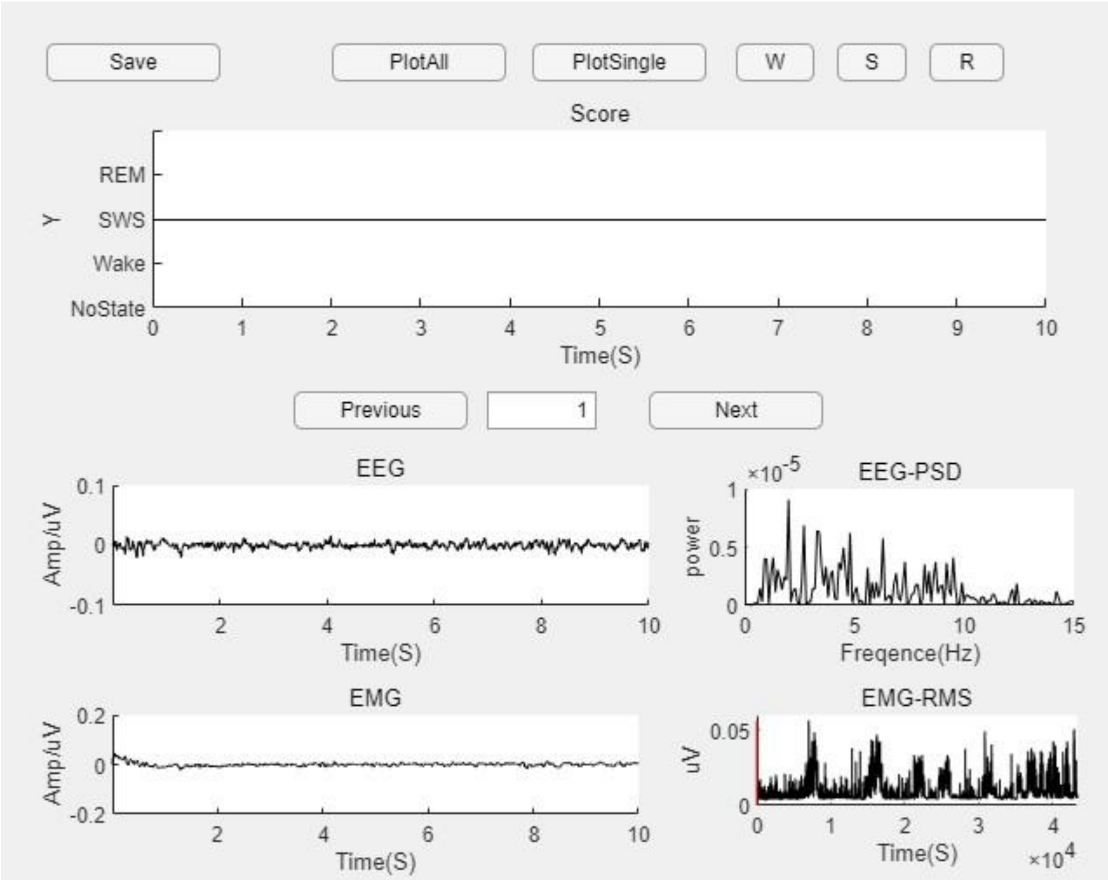
If certain staging results are unsatisfactory, select **fine adjustment** mode.

Click **[PlotSingle]** to examine:

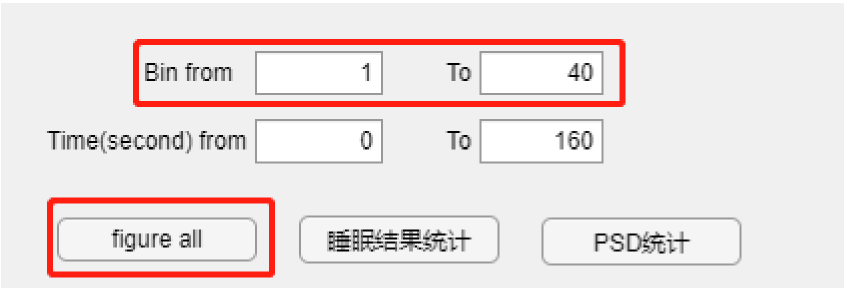
- EEG waveform
- EEG power spectral density (EEG-PSD)
- EMG waveform
- EMG RMS (red line indicates the current epoch)

Use:

- **[Previous] / [Next]** to navigate between epochs
- **[W] / [S] / [R]** to change the sleep state



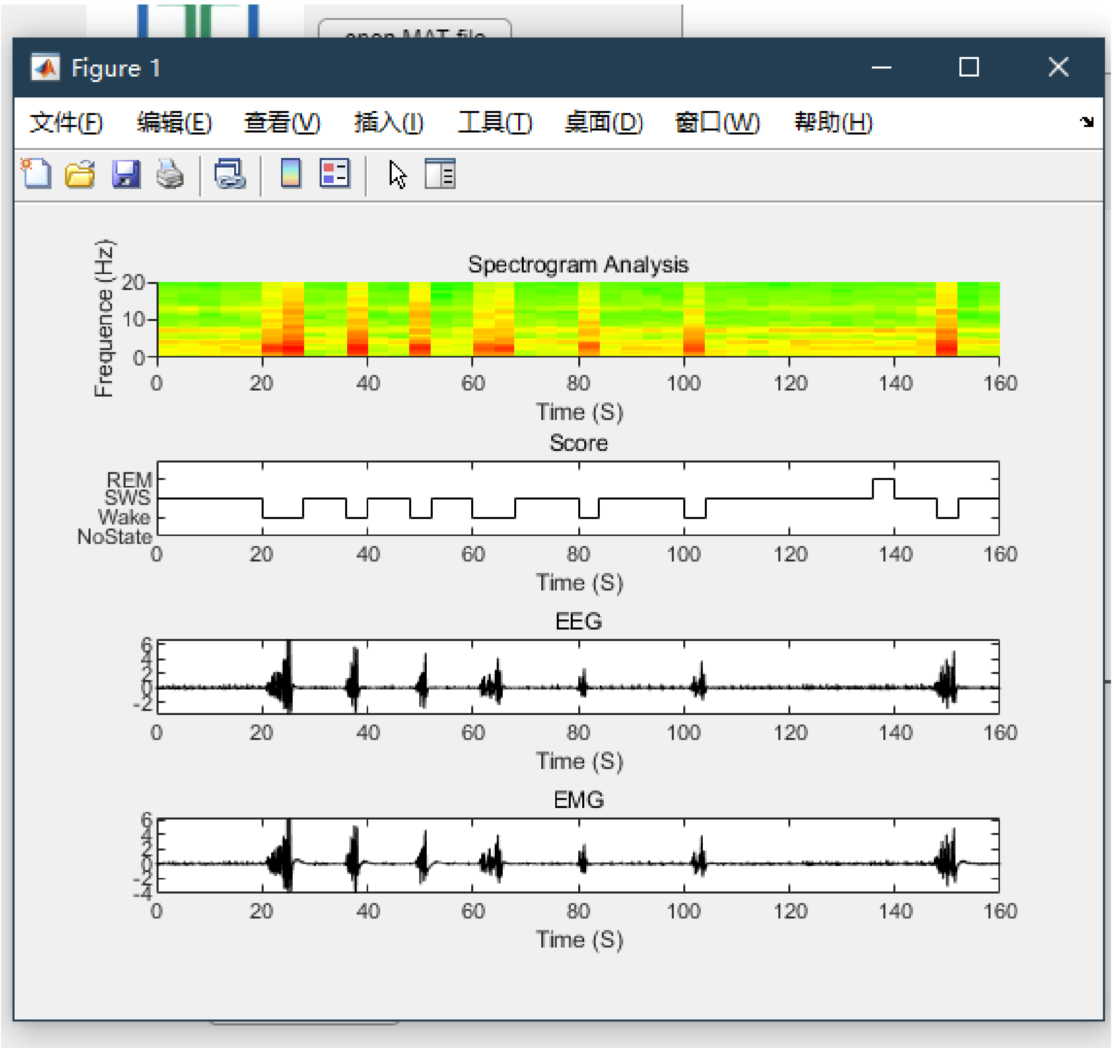
2.7 Analysis and Visualization of Results

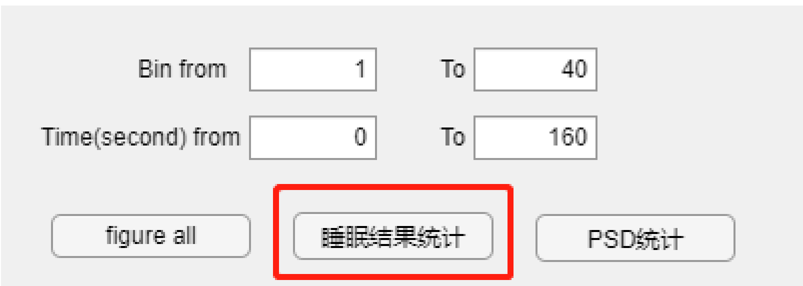


Select the **bin range**, then click **[Figure All]** to generate:

- Spectrograms
- Sleep score plots
- EEG plots
- EMG plots

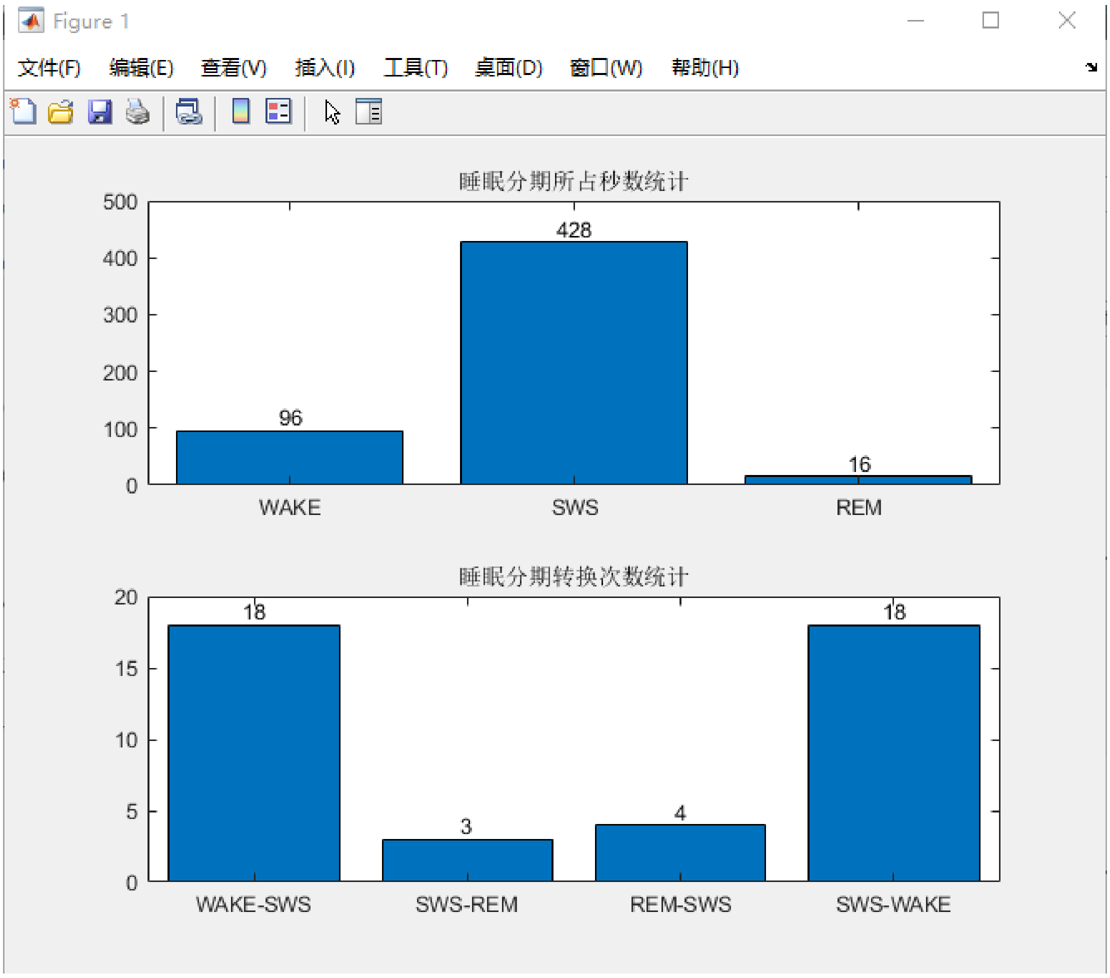
for the selected time range.



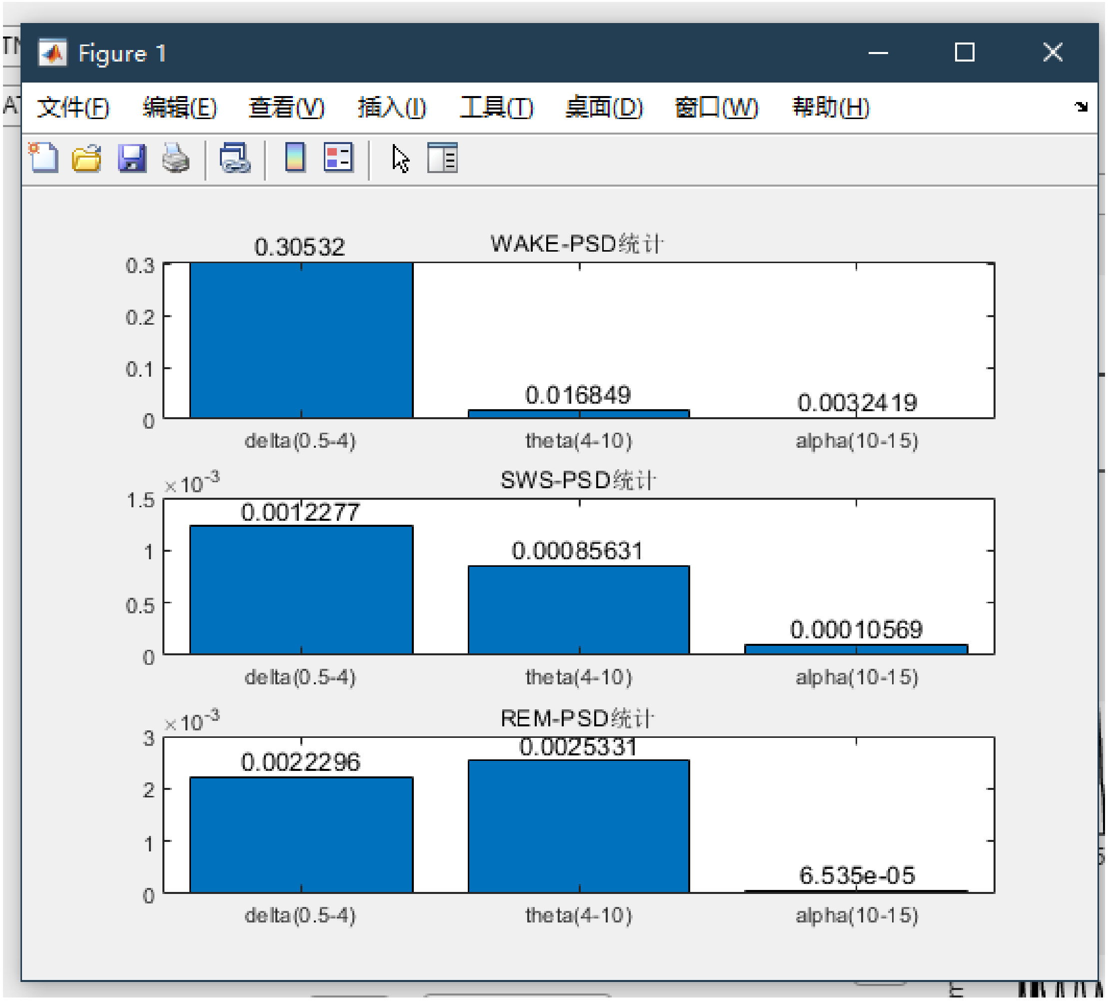


Click **Sleep Result Statistics** to obtain summary statistics.
The output includes:

1. A bar plot showing the total duration (in seconds) of **WAKE**, **SWS**, and **REM** within the selected time range
2. A plot showing the number of transitions:
   - WAKE → SWS
   - SWS → REM
   - REM → SWS
   - SWS → WAKE



Click **PSD Statistics** to obtain power statistics of different frequency bands under **WAKE**, **SWS**, and **REM** states.

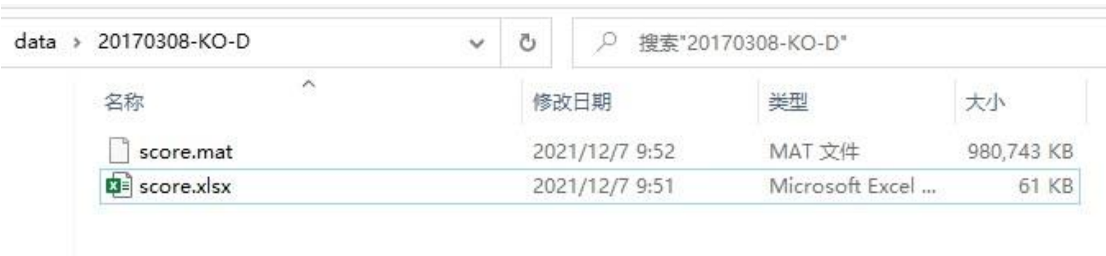


2.8 Saving Sleep Scoring Results

Click **[Save]** to save the results.

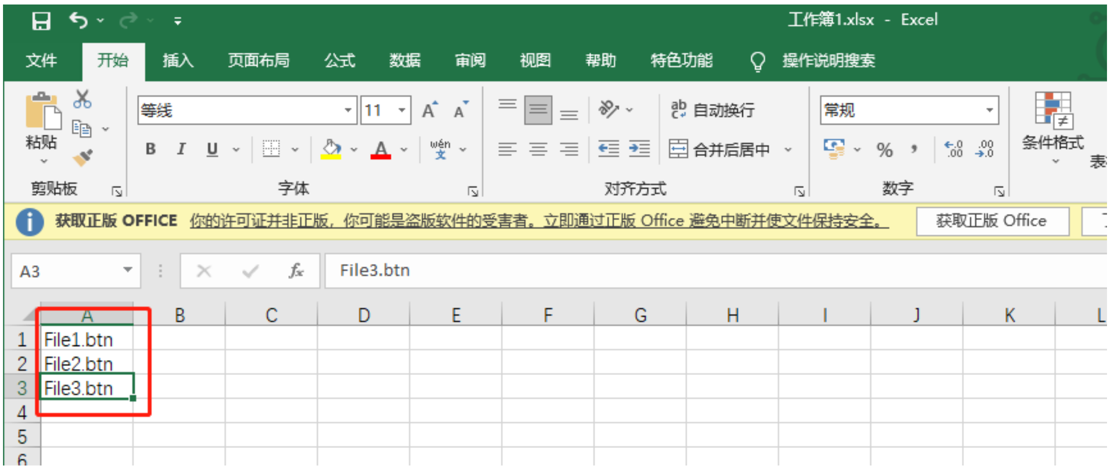
The results are saved in the same folder as the original data, generating two files:

- **score.mat**
  Contains the original data, extracted features, and sleep scoring results, facilitating future extensions of scoring functionality.
- **score.xlsx**
  Contains the exported sleep scoring results in spreadsheet format.

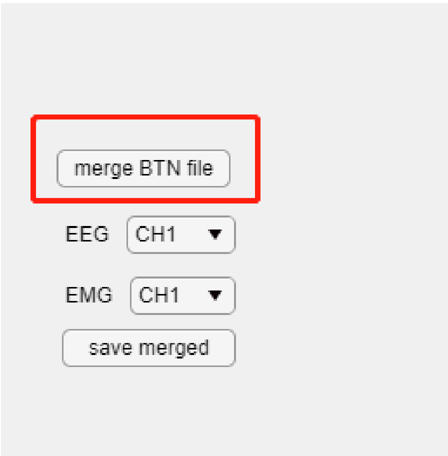


2.9 Merging BTN Files

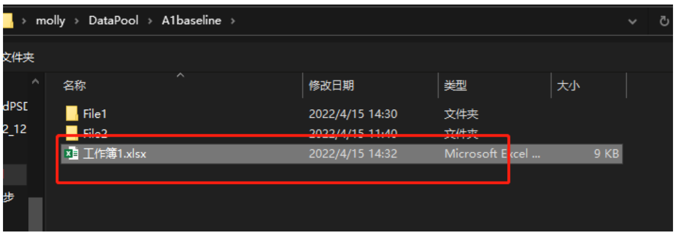
Create an **Excel file** and list, in order, the names of the data files to be merged.

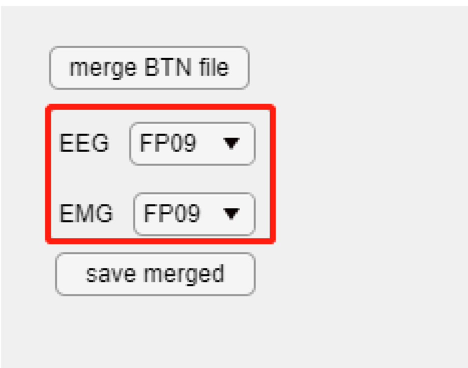


Click **Merge BTN File** and select the previously created Excel file.



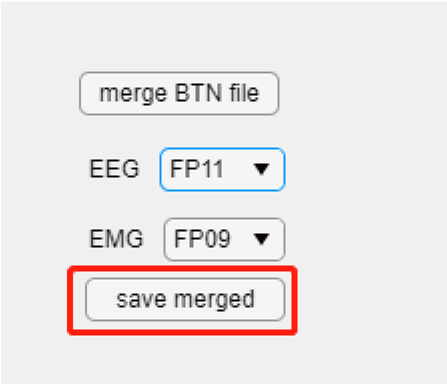
Select the corresponding **EEG** and **EMG** channels.





Click **Save Merged** to save the merged file.

⚠️ *Note: When the data size is large, saving may take some time.*



A file named **merge.mat** will be saved in the original data folder.

Click **Open MAT File** to open the merged

MAT-format file.